Indranil Paul ↗

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RESEARCH INTERESTS

Probabilistic machine learning, Bayesian deep learning, uncertainty quantification, deep generative models (VAEs, diffusion models, and normalizing flows), geometry of learned representations and latent spaces, Gaussian processes, kernel methods, and the theoretical foundations of deep learning. I am also interested in training-data attribution and data-centric AI. More broadly, I aim to develop mathematically principled methods connecting probabilistic inference with modern deep learning to build interpretable, and computationally efficient AI systems.

EDUCATION

MTech, Computer Science & Engineering — IIT Kanpur *2023 – 2025*

CGPA : 8.57 / 10

Thesis : **Retrain-Free Influence Estimation in Deep Neural Networks via Neural Tangent Kernel and Representer Theorem**

Extended as : "A Tale of Two Readouts: Unmasking Data Attribution through a Unified Kernel Framework" (Decision pending for TMLR)

Supervisor : Prof. Piyush Rai

Relevant Coursework: Probabilistic Machine Learning (CS772), Probability Theory for Computer Science (CS798F), Mathematics for Computer Science (CS601), Introduction to Machine Learning (CS771), Deep Reinforcement Learning (CS780), along with Deep Generative Models, Linear Algebra, and Stochastic Methods.

BTech, Computer Science & Engineering — Government College of Engineering & Leather Technology, Kolkata *2012 – 2016*

CGPA : 8.18 / 10

RESEARCH OUTPUT

A Tale of Two Readouts: Unmasking Data Attribution through a Unified Kernel Framework

Indranil Paul, Piyush Rai *Decision pending for TMLR / 2026 /* *OpenReview*

- Developed **NTK-JL**, a retrain-free training-data attribution framework that unifies prediction decomposition and counterfactual sensitivity through a shared empirical **Neural Tangent Kernel (NTK)** perspective.
- Established the connection between the two readouts: the decomposition readout recovers **Representer Point Selection (RPS)** in the frozen-backbone regime, while the datamodel readout connects to single-model **TRAK** under matched projected-gradient and Gauss–Newton approximations. The kernel ridge regression formulation admits a Gaussian-process posterior-sensitivity interpretation.
- Evaluated attribution fidelity across ViT-B/16, ResNet-50, and RoBERTa; achieved Spearman $\rho = 0.956$ on ViT-B/16 in the frozen-backbone setting. Head-only datamodel-margin experiments achieved **LDS 0.63 / 0.75** on SST-2 / QNLI.
- Improved scalability through **Johnson–Lindenstrauss sketching** and parameter subsampling, reducing memory requirements while preserving useful attribution fidelity. Also demonstrated the utility of self-influence scores for identifying potential annotation errors.

INDUSTRY EXPERIENCE

Software Engineer — Airawat Research Foundation, Kanpur *Jul 2025 – Mar 2026*

- Developed time-series ML pipelines for PM2.5 source apportionment and airshed boundary detection over IoT sensor streams from 100+ monitoring stations across Uttar Pradesh and Bihar.
- Fine-tuned **LLMs** using supervised fine-tuning, preference alignment, and PEFT for automated environmental analytics.
- Applied statistical modelling and exploratory data analysis to improve data quality and support environmental monitoring.

Data Intelligence Engineer / Analyst — EMC2, Bengaluru *Feb 2017 – Aug 2022*

- Built 30+ **ETL/ELT pipelines** (Informatica, Python) processing **10M+ records/day**; developed

Tableau dashboards reducing reporting latency by **35%**.

TEACHING EXPERIENCE

Teaching Assistant IIT Kanpur

- Teaching Assistant for **CS772: Probabilistic Machine Learning** (Prof. Piyush Rai).
- Teaching Assistant for **CS771: Introduction to Machine Learning** (Prof. Piyush Rai).
- Teaching Assistant for **ESC101: Fundamentals of Computing**
- Evaluated assignments, examinations, and student coursework.

SELECTED PROJECTS

Semantic Book Recommendation using LLMs | GitHub *Sep 2025*

Python, Transformer Embeddings, ChromaDB, BART-MNLI, RoBERTa, Gradio

- Built an end-to-end semantic retrieval and recommendation system over 5.2K+ books using Transformer embeddings and ChromaDB vector search (sub-second latency), with a multi-stage ranking pipeline combining zero-shot genre normalization and emotion-aware reranking.

Inductive Biases in Vision Models: CNNs vs. Vision Transformers | GitHub *May 2024*

PyTorch, EfficientNet-B0/B2, ViT-Base, Transfer Learning

- Empirically studied inductive bias and data efficiency of CNNs and Vision Transformers on FoodVision Mini: EfficientNet-B0/B2 achieved 83–85% accuracy with 60% less training, while **ViT-Base** reached **93%+ accuracy** (+5–7% F1), demonstrating superior generalisation under limited-data regimes, consistent with the weaker spatial inductive bias and richer pre-trained representations of attention-based architectures.

PROFESSIONAL SERVICE

- Reviewer: **ICLR 2026 Interpretability & Explainable AI Track** and **2027**

HONORS & AWARDS

- GATE CS: **AIR 156 (2023)**, **AIR 390 (2021)** ∼100,000 candidates
- **AIR 7**, ISI Kolkata MTech CS Entrance Examination, 2022

TECHNICAL SKILLS

Languages	Python, C/C++, SQL
Theory	**Probabilistic & Bayesian ML**, **Probabilistic Generative Models** (VAEs, GANs, Diffusion, Flows), **Neural Tangent Kernel (NTK)**, **Gaussian Processes**, **Influence & Attribution**, **Randomized Linear Algebra** (Sketching), **Variational Inference**
Deep Learning	**Transformers** (Encoder-only, Decoder-only), **Vision Transformers (ViT)**, CNNs, Transfer Learning, Fine-tuning
Frameworks	**PyTorch**, Hugging Face Transformers, Scikit-learn, NumPy, Pandas, Matplotlib, Manim
LLM Systems	**SFT**, **PEFT**, **DPO**, RAG Pipelines, LangChain
Tools	Git, WandB, Jupyter, VSCode, LaTeX

REFERENCES

Prof. Piyush Rai

Thesis Supervisor

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Prof. Sumit Ganguly

Graduate Coursework

Dept. CSE, IIT Kanpur

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